

Enhancing Well-being Through Food: A Conversational Agent for Mindful Eating and Cooking

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ABSTRACT

This study explores the potential of conversational agents to improve well-being during food preparation and consumption. We describe the process of designing and evaluating a conversational agent that promotes mindful eating and cooking. We first discuss the findings of an exploratory study conducted to understand changes in food behaviors of young adults in Mexico during the COVID-19 pandemic and their impact on well-being. Subsequently, an iterative methodology is used to design and evaluate two interactive systems with 2 different studies. The results of these interventions provide evidence that the systems improve the eating/cooking experience and promote mindfulness when performing these activities.

CCS CONCEPTS

• **Human-centered computing** → *Ubiquitous and mobile computing systems and tools.*

KEYWORDS

Conversational Agents, Mindful Eating and Cooking, Human Food Interaction

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1 INTRODUCTION

The recent COVID-19 pandemic considerably impacted eating and cooking patterns, including some negatively affecting well-being. For instance, isolation and people's anxiety increased negative eating behaviors like overeating which is linked to binge eating disorder [3]. In this scenario, nutritional experts recommend mindfulness during meal times to encourage healthy decisions [4]. Mindful eating is characterized by being fully present during meals, attentively

observing how food affects our senses, and acknowledging the physical and emotional sensations that arise while eating [18]. It has been proved that this technique positively impacts people's eating behavior [7].

Currently, existing applications utilize mindfulness techniques to assist people with meditation (e.g., [headspace.com](https://www.headspace.com)). While these apps facilitate general-purpose meditation, they are not specifically geared toward mindful eating/cooking. These applications only provide general mindful audio guides and lack an understanding of the users' activity and its situation. According to research, conversational agents can encourage users to have long interactions and enhance their experiences which is partly due to the more natural and conversational style of the interaction provided [10].

In the literature, there are prototypes centered on mindfulness, but not necessarily around food. For example, Inner Garden [16], a sandbox that combines augmented and virtual reality to provide a mindful interactive experience where users can directly interact with the sand. A few prototypes have been proposed for mindful eating. For instance, an intelligent spoon helps people focus on their food by dropping it if they become distracted [9]. Even though there are a few examples of technological solutions that promote mindfulness, the food experience, and especially the cooking experience has yet to be explored.

This paper aims to improve people's well-being by designing interactive systems that promote healthy food behaviors like mindful eating/cooking. To achieve this, it was crucial to understand the current eating experience among individuals and identify potential design opportunities in this context. Thus, we first describe insights from an exploratory study conducted with young adults in Mexico. After that, we present the procedure to design and implement interactive technologies with the purpose of improving the food experience with a mindfulness approach.

2 RELATED WORK

A conversational agent or chatbot is a program crafted to replicate human-like conversations with users [2]. These agents have widespread applications in various domains, including education [19] and healthcare [17]. Prototype conversational agents have also been explored in nutrition, for example, "Lucy" [15], a digital assistant designed to help users lose weight by monitoring their eating behaviors, and others [8] showcase their ability to adapt dialogue characteristics based on users' emotional cues.

In the pursuit of developing technologies that promote well-being, a crucial aspect involves gaining a comprehensive understanding of individuals' eating behaviors. Some studies [11, 12] have been dedicated to identifying and studying these behaviors,

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highlighting the importance of social factors in food consumption patterns.

There have been a few recent studies exploring the use of conversational agents for mindful eating. For instance, the research conducted by [20] explores the intersection of mindful eating and conversational agents. Their study showcases the positive impact of these assistants in promoting healthier eating habits and fostering mindful consumption. To our knowledge, however, supporting mindful cooking through conversational agents has not been explored by the HCI/UbiComp community.

3 METHODOLOGY

Figure 1 illustrates the methodology used in this research, which is a user-centered design process [1]. After the exploratory study, an iterative process is conducted to design, implement, and evaluate interactive technologies that can enhance people’s food experience at home. We next describe each stage of the methodology.

- **Exploratory Study:** This process includes conducting virtual interviews with individuals to learn about their experiences with food at home. By analyzing qualitative data, we gain insights into people’s current cooking and eating habit in their own homes.
- **Design Sessions:** Based on the insights gained from the exploratory study, engage in collaborative design sessions. This step involves brainstorming, ideation, and sketching potential solutions to address the identified needs and enhance individuals’ food experiences. Encouraging the participation of designers, researchers, and relevant stakeholders to generate diverse ideas and perspectives.
- **Technology Implementation:** Transforming the selected design concepts into prototypes of interactive systems. Using appropriate tools and technologies to create systems that can improve individuals’ food experiences.
- **User Evaluation:** Conducting user testing and evaluation sessions to gather feedback on the implemented systems.
- **Results & Reports:** Analyze the data collected during the user evaluation phase. Identifying patterns, trends, and insights to gain a comprehensive understanding of the strengths and weaknesses of the interactive systems.

After completing the initial iteration from step 2 to step 5, the process returns to step 2, the Design Sessions. This allows for the integration of user feedback, and the exploration of new ideas and possibilities. The iterative nature of this loop ensures that the interactive systems continually evolve and improve to meet individuals’ changing needs and preferences in their food experiences at home.

4 EXPLORATORY STUDY AND SCENARIOS

A qualitative study was conducted to understand individuals’ eating and cooking behaviors at home. The study started in 2020 during the COVID-19 pandemic, which allowed us to highlight instances of people’s food experiences isolated in their homes. For this purpose, 14 young professionals with an average age of 26 were recruited, and semi-structured interviews were conducted covering topics related to lifestyle, eating and cooking behaviors, and technology usage.

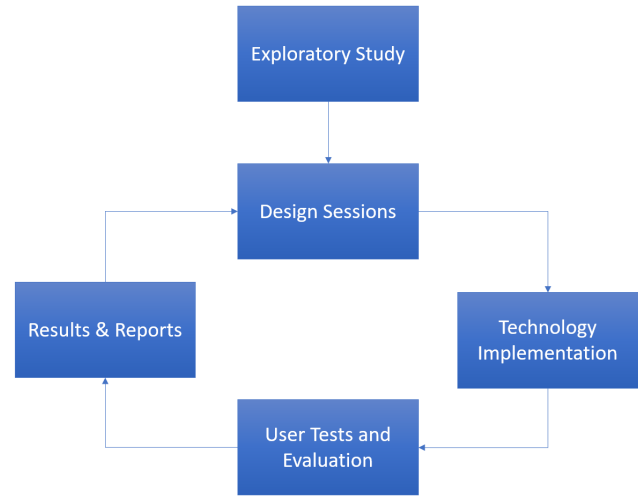


Figure 1: Iterative, user-centered methodology followed in the research.

The interviews were conducted through video conferencing and had an average duration of 36 minutes (± 11.5 minutes). Subsequently, a line-by-line coding process was conducted to extract different themes from the participants’ responses. The following are the four main themes derived from the interviews [13] 1) negative feelings were impacting eating patterns (e.g., binge eating), 2) some found the pandemic as a motivation to improve their health, 3) increased home cooking and eating, and 4) reduced food socialization while eating and cooking.

This study helped us understand how the pandemic changed people’s eating behaviors, mostly negatively. For example, the stress and anxiety caused by the pandemic led some people to engage in negative behaviors like overeating or binge eating. Similarly, isolation caused anxiety in others due to being away from their loved ones. On the other hand, some people took this opportunity of staying at home to cook their meals instead of ordering food or eating out. These behaviors were triggered by the confinement situation, but it helped highlight situations to understand how people eat at home regularly.

4.1 Scenarios for Design

The four main themes derived from the qualitative study helped identify different scenarios that illustrate challenges people were currently experiencing. Based on these scenarios, design sessions were conducted. Two virtual sessions were held with individuals possessing expertise in technology: the first session involved three software designers/programmers, and the second session involved two informants that participated in the interviews. We have created 2 scenarios and identified potential solutions for each.

Scenario 1: Sofia is a new graduate student studying away from her hometown and living alone in an apartment near her school. She has been feeling lonely for being away from her loved ones and stressed due to adapting to a new life away from home and a demanding study schedule. These circumstances have led to changes

in Sofia's behavior such as frequently ordering takeout instead of cooking her meals. Additionally, she tends to eat compulsively and in large quantities, resulting in rapid weight gain. As a result, Sofia sought help from a nutritionist to improve her eating habits. After hearing about her situation, the nutritionist suggested that Sofia incorporate mindful eating techniques to nourish herself and start cooking her meals. However, Sofia found it challenging to focus on her meals as she often got distracted watching television while eating. This led the nutritionist to recommend a technological solution called MyndFood, a voice assistant that utilizes mindfulness techniques to guide people during cooking and eating. Following the nutritionist's advice, Sofia now uses this voice assistant every day during her meals alone in her apartment. Although she has only been using it for two weeks, Sofia enjoys her meals much more and consumes the right portions. The soothing guidance from the voice assistant helps her relax and release the day's stress. Similarly, the voice assistant has helped Sofia improve her eating behaviors and maintain a healthier weight. She tends to cook more meals at home because she appreciates the interaction with the assistant and has access to various recipe options that assist her in preparing her meals better.

Scenario 2: Susy has been feeling isolated and lonely due to the lack of social interactions because of the confinement. She misses the office banter and the casual conversations with colleagues that used to brighten up her day. The absence of companionship during mealtimes has also made her meals less enjoyable and more monotonous. One day, Susy stumbled upon an ad on social media for an innovative solution called "TableFor3". This mobile application connects 3 individuals who are dining alone by allowing its users to enter a virtual dining table and engage in conversations with other users while enjoying their meals from their homes. Susy downloaded the "TableFor3" app and created her profile. She customized her preferences and interests to ensure compatibility with potential dining companions. Whenever Susy was ready to have a meal, she would open the app, select her preferred time slot, and wait for a matching virtual friend. During mealtimes, Susy would connect with her dining friends through the app's audio and video chat feature. They would share stories, discuss common interests, and even exchange cooking tips and recipe ideas. Susy found that having someone to talk to during meals made her feel less lonely and more connected to the outside world. The conversations were not only enjoyable but also provided a welcome distraction from work-related stress.

After developing the scenarios with their potential technological solutions, sessions were held with the researchers, who are the authors of this work, to decide on the feasibility and potential impact of developing one of these technologies. From this, Scenario 1 was chosen for further development, and work began on designing and implementing a conversational agent that utilizes mindful eating techniques to enhance the eating experience.

5 STUDY 1: A MINDFUL TEA EXPERIENCE

In this study, we conducted an experiment to assess the effectiveness of a smart environment designed to enhance the pleasurable experience of mindful eating and drinking. The core component of this environment, called CATE, is a conversational agent that serves

as a central hub, guiding users in the process of preparing and savoring tea along with a snack. Here, we wanted to compare how participants perceived the system under two different conditions: mindfulness and music.

Figure 2 shows the setup for the experiment [14]. The whole system incorporates various elements and devices, such as a Smart Box with teas and snacks, a Conversational Agent Hub for audio interaction and instructional images, a Window View with a simulated scenic background, a Lamp for better visibility, and a Kettle with LED lights. For this prototype, a Wizard of Oz approach is used, where pre-recorded sentences are played by a Wizard who monitors interactions and controls devices remotely to ensure a smooth experience without system bugs or network delays.

Users engage with the agent during their tea-drinking and snack-eating sessions. The system initiates small talk by asking questions about the participant's well-being and stress levels to provide a comforting experience. Additionally, following a mindful approach, the system guides the user through the tea-tasting process, encouraging the use of all senses. It prompts the user to identify the tea's color, smell it, and then taste it, repeating a similar pattern with the snacks. Relaxing background music complements the experience.

To conduct the study, we recruited two groups of independent participants, with 10 individuals in the mindful condition and 12 individuals in the music condition. The participants in the mindful condition interacted with the conversational agent that focused on promoting mindfulness during their tea-drinking experience, while the participants in the music condition were exposed to an agent that played background music instead.

The results indicated that participants who engaged with the mindful conversational agent rated the system more favorably with respect to likeability compared to those who interacted with the music-playing agent. In the mindful condition, participants reported having a more joyful experience with the system, suggesting that the mindfulness-focused agent had a positive impact on their overall experience. Furthermore, participants in the mindful condition expressed an increased appreciation for the visual aspects (colors) and olfactory sensations (smells) associated with their tea. This finding suggests that the mindful agent successfully facilitated a heightened awareness of the sensory elements involved in the tea-drinking experience, enhancing the participants' overall enjoyment.

6 STUDY 2: A MINDFUL SUSHI COOKING EXPERIENCE

For this experiment, we decided to expand more on food preparation to gather insights about the impact of mindfulness on peoples' food experiences. We chose sushi since it provides a variety of ingredients and is relatively easy and quick to prepare even for novice cooks. Figure 3 shows the setup for the second study where a conversational agent is used to prepare and taste sushi. For this study, the agent is controlled through the Wizard of Oz method for navigation. The following are the main characteristics of the conversational agent:

- **Recipe:** It provides a complete guide with the necessary steps to prepare sushi, along with having all the required ingredients in the scenario.

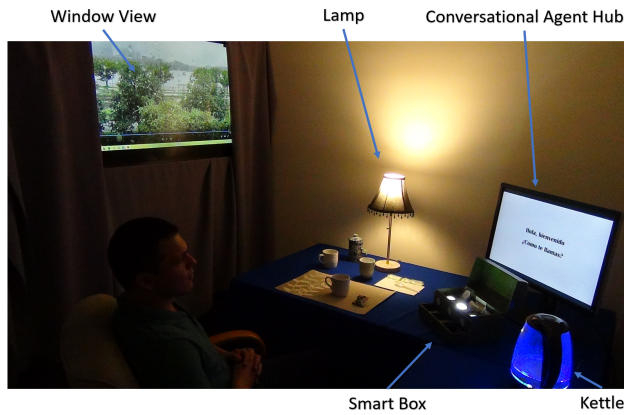


Figure 2: Study 1 Setup

- **Mindfulness:** It offers participants the option to engage in a mindful interaction, guiding them through the steps to use all their senses and fully concentrate on the food to enhance their experience. (This feature is optional)
- **Multimodality:** It utilizes various media to provide feedback to participants, including audio and text displayed on a screen. It also employs videos to enhance the explanation of certain food preparation steps.
- **Navigation (Wizard of Oz):** It understands voice commands commonly used by conventional conversational agents, such as "next," "back," "repeat," and "ready," among others.



Figure 3: Study 2 Setup

A total of 40 people participated in the study, 30 women and 10 men, and we divided participants into two groups. The first group interacted with a mindful conversational agent that encourages users to involve all senses while cooking. And a second group with participants who interacted with a non-mindful or more conventional voice agent with the necessary instructions for sushi preparation. After each session, participants had the opportunity to comment about their experience with the agent, and we measured consciousness level [6] and enjoyment of the system [5].

Preliminary results for this study revealed that participants who engaged with the mindful conversational agent exhibited a greater level of consciousness and mindfulness during their food experience compared to participants who interacted with the non-mindful agent. This suggests that the mindful agent successfully facilitated a state of awareness among users, enabling them to fully immerse themselves in the present moment while preparing food and engaging with the voice agent.

Moreover, participants who interacted with the mindful agent not only reported a higher level of awareness but also expressed an overall more positive experience in their interactions with the voice agent compared to the non-mindful group. This finding implies that the incorporation of mindfulness elements in the agent's dialogue and guidance positively influenced the users' perception of the interaction, enhancing their overall satisfaction and enjoyment.

6.1 Participants' comments

The participants of the study provided valuable insights about the conversational agent that guides the cooking and eating experience. The most common characteristic mentioned by participants was a pleasant experience. This suggests that the agent created a positive and enjoyable experience for the users. Similarly, another frequently mentioned characteristic was that the agent was likable. This suggests that the agent was able to establish rapport with the users and was perceived as a friendly dialogue system. Additionally, the agent was also perceived as easy to use, which suggests that the agent was successful in simplifying the cooking process for the users.

Furthermore, participants who interacted with the mindful agent mentioned that the agent made them feel relaxed. This indicates that the agent was able to create a calm and relaxing experience for the users. In a similar manner, participants mentioned that mindfulness was very important. This indicates that the conversational agent created a mindful experience for the users, making the experience more meaningful and purposeful.

7 DISCUSSION

The results of the experiments provide the foundation for future research in the design and implementation of technologies that enhance the food experience. However, one key aspect for the development of future prototypes is customization. For example, while the majority of participants enjoyed interacting with a mindful conversational agent in Study 2, it was noted by one participant that it occasionally caused distraction during cooking, and another participant expressed dissatisfaction with the agent's voice.

Additionally, although a Wizard of Oz technique was employed in the prototypes implemented for these experiments to mitigate bias in technological errors, it would be advantageous to automate the prototypes and validate them outside of laboratory settings. Moreover, the incorporation of features using activity recognition models for cooking tasks such as chopping ingredients or detecting moments when the user tastes their food would improve the timing and type of feedback from the system.

8 CONCLUSIONS AND FUTURE WORK

We have described two studies conducted with the aim of enhancing the food experience with mindfulness using a conversational agent. Both studies involved comparing two different approaches of agents to assess the effectiveness of mindfulness. Overall, the results indicated that participants enjoyed the experience more when using a mindful agent while cooking and tasting the food, compared to the non-mindful agents.

These findings suggest that a mindful conversational agent can enhance users' experience in food preparation and tasting. Furthermore, these studies could lay the groundwork for the development of more advanced and mindful voice assistants that can improve our relationship with food and technology.

For future work, an autonomous version of the agent will be implemented to assist with food preparation and consumption. We aim to explore the use of Large Language Models (LLM) and activity recognition to adapt the agent's behavior based on the person's tasks. For instance, the agent will be able to recognize when the person is cutting ingredients and provide relevant recommendations. Moreover, with the emergence of new tools like chatGPT (<https://chat.openai.com>), we expect to enhance the agent's dialogue interaction with the user. We will experiment with different types of prompts and personalities tailored to specific food activities or scenarios.

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